

Sound

SOUNDS ARE VIBRATIONS, CARRIED EITHER BY SOLIDS, LIQUIDS, OR GASES.

SEE ALSO

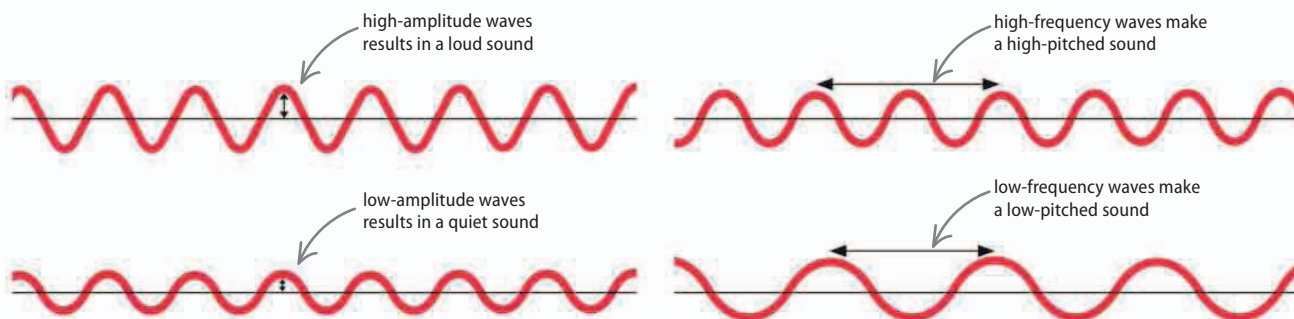
- ◀ 64 Hearing
- ◀ 184 Atmospheric pressure
- ◀ 192–193 Waves

Sounds are of great benefit for communication and can also be harnessed for medical or industrial use. However, unwanted sound is a serious pollutant that damages health and well-being.

Pitch and loudness

The characteristics of sound that we experience as pitch and loudness are closely related to the physical properties of sound waves. Generally, the higher the frequency of a wave—the number of peaks and troughs that pass a point each second—the higher its pitch; the larger the amplitude of the wave, the louder it sounds.

We can hear sounds so quiet that they make our eardrums move **less than the width of an atom**.



△ Loudness

These sound waves have the same frequency but different amplitudes. A higher amplitude indicates there is a larger variation in air pressure, and greater volume.

△ Pitch

These sound waves have the same amplitude but different frequencies. A higher frequency creates a more rapid variation in air pressure and results in a higher pitch.

Echoes

Sound waves reflect from surfaces, especially hard, smooth ones. If the surface is far enough from the sound source for an adequate time to pass before the reflection returns, it can be heard or detected as an echo. Underwater echoes are used by ships to scan the sea floor. The return time depends on the depth of the bed, so maps of the seafloor can be made in this way.

▷ Mapping the seabed

This diagram illustrates how a ship uses echoes to map the sea floor.

